

Industrial CO2 Emission Prediction Report

Report Generated: 17-09-2026 12:05:12
Record Timestamp: 17-09-2026 12:04:59
Prediction Record ID: #56

Core Telemetry Parameters

Parameter	Observed Value
Energy Consumption	163.00 kWh
Production Count	233.00 Units
Working / Shift Hour	15:00 hrs
Manufacturing Schedule	Standard Weekday

Emission Forecast & Analysis

Forecasted CO2 Emission	133.66 kg CO2
Pollution Classification	Medium
Inference Methodology	Electricity Emission Factor (0.82 kg CO2/kWh)
Grid Emission Constant	0.82 kg CO2/kWh

Recommended Mitigation Measures

- Investigate sub-optimal machine operating loads.
- Optimize machine operating hours and idle intervals.
- Perform preventive maintenance on high-draw equipment.
- Improve production throughput per kilowatt-hour.
- Review power factor and harmonic distortion levels.
- Schedule heavy processing during off-peak tariff periods.

This certification report was generated by the Industrial Carbon Emission Forecasting System. Telemetry input is evaluated using Scikit-Learn Machine Learning models and Scope 2 emission standards.